

which produces the following output:

L^AT_EX can now compute: $\frac{\sin(3.5)}{2} + 2 \cdot 10^{-3} = -0.1733916138448099$.

`\interval {<integer expression>}`

The expandable command `\interval` takes as its argument an integer expression and produces a result using the normal rules of mathematics with some restrictions, see below. The operations recognized are `+`, `-`, `*` and `/` plus parentheses. As this command is expandable it can be used where T_EX requires a number and for example within a low-level `\edef` operation to give a purely numerical result.

This is basically a thin wrapper for the primitive `\numexpr` command and therefore has some syntax restrictions. These are:

- `/` denotes division rounded to the closest integer with ties rounded away from zero;
- there is an error and the overall expression evaluates to zero whenever the absolute value of any intermediate result exceeds $2^{31} - 1$, except in the case of scaling operations `a*b/c`, for which `a*b` may be arbitrarily large;
- parentheses may not appear after unary `+` or `-`, namely placing `+(` or `-(` at the start of an expression or after `+`, `-`, `*`, `/` or `(` leads to an error.

An example of use could be the following.

L^AT_EX can now compute: The sum of the numbers is `\interval{1 + 2 + 3}`.

which results in “L^AT_EX can now compute: The sum of the numbers is 6.”

`\dimeval {<dimen expression>}      \skipeval {<skip expression>}`

Similar to `\interval` but computing a length (`dimen`) or a rubber length (`skip`) value. Both are thin wrappers around the corresponding engine primitives, which makes them fast, but therefore shows the same syntax peculiarities as discussed above. Nevertheless, in practice they are usually sufficient. For example

```
\NewDocumentCommand\calculateheight{m}{%
  \setlength\textheight{\dimeval{\topskip+\baselineskip*\interval{#1-1}}}}
```

sets the `\textheight` to the appropriate value if a page should hold a specific number of text lines. Thus after `\calculateheight{40}` it is set to 478.0pt, given the values `\topskip` (10.0pt) and `\baselineskip` (12.0pt) in the current document.